



What Drives Restaurant Performance on Yelp: A Data-Driven Analysis

This project analyzes key factors driving restaurant performance on Yelp, focusing on how user credibility, pricing, and location influence ratings.

Key takeaway: while measurable factors such as elite reviewers have a positive impact, customer perception and experience are the primary drivers of ratings in driving ratings.



Part I: Linear Regression

Residuals:

Min	1Q	Median	3Q	Max
-4.7119	-0.4773	0.0815	0.5679	1.8011

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.354e+00	3.332e-02	100.649	< 2e-16	***
elite_cnt	2.393e-03	1.990e-04	12.026	< 2e-16	***
price_levelprice_2	2.444e-01	2.052e-02	11.908	< 2e-16	***
price_levelprice_3	2.084e-01	5.851e-02	3.561	0.000372	***
price_levelprice_4	4.913e-01	9.273e-02	5.298	1.21e-07	***
biz_age	-8.795e-05	7.396e-06	-11.893	< 2e-16	***
metroPhoenix_area	1.043e-01	3.081e-02	3.387	0.000710	***
metroPittsburgh	1.291e-01	4.114e-02	3.137	0.001714	**
metroVegas_area	8.726e-02	3.221e-02	2.709	0.006770	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7605 on 5957 degrees of freedom
Multiple R-squared: 0.07706, Adjusted R-squared: 0.07582
F-statistic: 62.17 on 8 and 5957 DF, p-value: < 2.2e-16

Overview

- **P-values:** All variables statistically significant (< 0.05), confirming each variable has an effect on rating.
- **Coefficient estimates:** All variables have a positive effect on the rating, except for restaurant age.
- **Standard errors (SE):** Relatively small compared to the coefficients, indicating that the estimates are precise.
- **R-squared:** The model explains about 7.7% of the variation in restaurant ratings.

Interpret of the Elite

- The **coefficient** of elite_cnt (0.002393) represents that 1 unit increase in elite_cnt will increase 0.0024 in restaurant ratings (biz.stars), holding all other variables constant
- The effect is statistically significant, as the p-value is less than 0.001, which means Elite is influential on rating.

Restaurants with more elite reviewers receive slightly higher ratings.

Suggesting elite users enhance perceived quality and credibility.

Fraction of Variation

- **7.7% ($R^2=0.077$) fraction of variation in restaurant ratings is explained by the model**
- **The magnitude:** a relatively low R-squared, indicating that the model has limited explanatory power
- **Prediction Quality:** The model is not very strong in predicting restaurant ratings, as a large portion of variation remains unexplained.



Business Implication: While factors such as elite reviewers, price level, business age, and location are statistically significant, they do not capture most of what drives customer ratings. Other factors such as food quality, service experience, and brand perception, are likely more important.

Part 2: Model Comparison for Linear Regression

```
> anova(model1, model2)
Analysis of Variance Table

Model 1: biz.stars ~ elite_cnt
Model 2: biz.stars ~ elite_cnt + price_level + biz_age
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1   5964 3644.6
2   5960 3453.1  4    191.54 82.648 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> anova(model2, model3)
Analysis of Variance Table

Model 1: biz.stars ~ elite_cnt + price_level + biz_age
Model 2: biz.stars ~ elite_cnt + price_level + biz_age + metro + biz.rws.cnt
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1   5960 3453.1
2   5956 3366.8  4    86.292 38.163 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

1 VS. 2

- F statistic: 82.648
- P-value: <2.2e-16 (significant)
 - Reject the null hypothesis

Adding price and age of business improves the model



2 VS. 3

- F statistic: 38.163
- P-value: <2.2e-16 (significant)
 - Reject the null hypothesis

Adding Metro and total number of reviews improves the model



Implications

- **F-tests** indicate that each additional set of variables significantly improves the model, and Model 3 had the smallest RSS, so we chose **Model 3**.
- This means that restaurant ratings are influenced by multiple factors, including elite reviewers, price level, business age, location, and review volume.
- For the business, using a more comprehensive model like Model 3 allows for better understanding of what drives customer ratings and supports more informed decision-making.

Part 3: Logistic Regression: Predicting Business Survival

```
Call:
glm(formula = is_open ~ elite_cnt + price_level + biz_age + biz.rws.cnt,
     family = "binomial", data = biz.var.sample)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-9.587e-01	6.865e-02	-13.964	< 2e-16	***
elite_cnt	-4.597e-02	2.916e-03	-15.765	< 2e-16	***
price_levelprice_2	-5.872e-01	6.612e-02	-8.882	< 2e-16	***
price_levelprice_3	-1.235e+00	1.967e-01	-6.280	3.38e-10	***
price_levelprice_4	-9.097e-01	3.053e-01	-2.980	0.00288	**
biz_age	6.924e-04	2.717e-05	25.479	< 2e-16	***
biz.rws.cnt	1.127e-02	6.160e-04	18.296	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 7799.2 on 5965 degrees of freedom
Residual deviance: 6154.1 on 5959 degrees of freedom
AIC: 6168.1

Number of Fisher Scoring iterations: 6

Coefficient Estimates

The coefficients indicate that the number of Yelp Elite reviewers, price category 2, 3, and 4 are associated with a lower probability of business survival. In contrast, age of business and total number of reviews significantly increase the likelihood of a business remaining open.



Standard Errors

The relatively small standard errors across most predictors indicate high estimation precision, particularly for variables such as review count and business age. Higher price levels exhibit larger standard errors, suggesting greater variability within these categories.



P-values

All p-values are below 0.05, meaning that the relationship between the variables and the chances that a restaurant remains open are statistically significant



Deviance

The model demonstrates a substantial improvement in fit, with residual deviance decreasing from 7799.2 (null model) to 6154.1 after including predictors. This large reduction suggests that the independent variables collectively explain a significant portion of the variation in business survival.



Part 4: Converting Log-Odds to Odds Ratios

Number of Yelp Elite reviewers

Log-odds coefficient: -0.04597

The coefficient of -0.046 indicates that each additional unit increase in Yelp Elite reviewer count is associated with a statistically significant decrease in the likelihood of business survival. Specifically, it decreases the log-odds of a business remaining open by 0.046, holding all other variables constant.

Odds ratio: 0.955

A one-unit increase in elite reviewer count multiplies the odds that the restaurant remains open by 0.955 (corresponding to a 4.5% decrease in odds, holding other variables constant)



```
> exp(coef(m1))
      (Intercept)      elite_cnt price_levelprice_2 price_levelprice_3
      0.3834080      0.9550746      0.5558539      0.2907405
price_levelprice_4      biz_age      biz.rws.cnt
      0.4026621      1.0006926      1.0113351
```

Interpreting Price Level 2

- The odds ratio of **0.556** compares restaurants with price level 2 to those with price level 1 (the reference group)
- Restaurants with price level 2 **have 44.4% lower odds** of remaining open compared to price level 1
- This suggests that **mid-priced restaurants (price level 2) are less likely to survive compared to lower-priced restaurants**. One possible explanation is that they are not as affordable as budget options, yet may not offer the differentiation or perceived value of higher-end establishments. As a result, they may struggle more with customer retention and competitive positioning.

Part 5: Model Comparison for Logistic Regression

Likelihood Ratio Test: does adding variables meaningfully reduce prediction error, or just by chance?

Report on change in deviance, p-value, decision (Q10)

Comparison	Added Variables	Deviance	p-value
Model 1 → 2	Price Level + Business Age	948.29	< 2.2e-16, significant
Model 2 → 3	Review Count	513.73	< 2.2e-16, significant

Model 1 → 2 Decision: **Reject H0**, price level and business age significantly improve survival prediction beyond elite reviewers alone

Model 2 → 3 Decision: **Reject H0**, review count alone produces a massive improvement, confirming it is a key driver of restaurant survival on Yelp

What does a reduction in deviance mean? (Q11)

- Deviance = how poorly the model predicts restaurant survival; Lower deviance = better predictions
- Every variable added significantly lowers prediction error
- Review count alone drops deviance by 513 points, indicating the strongest single signal in the dataset

Business implication: Restaurants with lower price tiers tend to close more, while restaurants with longer operating history and more reviews are significantly more predictable in terms of survival. This means model 3 can give clients a much more accurate assessment by incorporating all three factors, rather than relying on elite reviewer activity alone.

Part 6: Model Selection

Model Specifications:

- Model 1: $\text{is_open} \sim \text{elite_cnt}$
- Model 2: $\text{is_open} \sim \text{elite_cnt} + \text{price_level} + \text{biz_age}$
- Model 3: $\text{is_open} \sim \text{elite_cnt} + \text{price_level} + \text{biz_age} + \text{biz.rws.cnt}$

Metric	Model 1 (1 predictor)	Model 2 (3 predictors)	Model 3 (4 predictors)
AIC	7,620.16	6,679.86	6,168.00
BIC	7,633.54	6,720.03	6,215.00

Statistical Interpretation:

- AIC drops by 940.29 from Model 1→2 and another 511.73 from Model 2→3, indicating substantial **improvement in predictive fit** at each step
- BIC penalizes complexity more heavily than AIC, yet **both agree on Model 3** — confirming the added predictors are justified and there is no trade-off between accuracy and parsimony

Business Implication

- Model 3 is recommended: in addition to review quality, overall review count likely reflects restaurant popularity and customer traction, which helps explain survival
- With only 4 predictors, the model remains simple, interpretable, and practical for business decision-making



Thank You!

